

# Computational Study of Carbon Dioxide Capture by Tertiary Amines

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The reaction mechanisms and corresponding structure-activity relationships of tertiary amines with respect to CO<sub>2</sub> capture have been investigated using density functional theory (DFT) calculations. The reaction mechanism for CO<sub>2</sub> capture via base-catalyzed hydration to form bicarbonate is proposed to proceed in a single step involving proton transfer and the formation of a carbon-oxygen bond. Based on the height of the reaction barriers, we suggest that amines containing side chains with the ethyl group, along with a single hydroxyl group, and cyclic structures, are especially active for CO<sub>2</sub> capture. The activation barrier is shown to be a descriptor for predicting the

experimental CO<sub>2</sub> loading values. To enhance the prediction accuracy for CO<sub>2</sub> loading, we employ the sure-independence screening and sparsifying operator (SISSO) method, which can scan a large pool of mathematical terms stemming from combining DFT-derived descriptors to select the superior ones. Thus, we can predict the CO<sub>2</sub> loading with acceptable accuracy from the obtained mathematical expression. Since the computational workload of applying this expression is negligible, this facilitates high-throughput screening and accelerates the design of tertiary amines for CO<sub>2</sub> capture.

## Introduction

The emissions of the greenhouse gas CO<sub>2</sub> over the past century are considered one of the main contributors to global warming. Thus, extensive efforts have focused on developing CO<sub>2</sub> capture and storage methods. Primary and secondary amines are widely recognized for their effectiveness in CO<sub>2</sub> absorption. However, the use of them still has some drawbacks, including volatilization and degradation, equipment corrosion, as well as the formation of thermally stable carbamate salts which require more energy for regeneration.<sup>[1]</sup> Aqueous tertiary amines offer advantages because they exclusively produce bicarbonates instead of carbamates in the process. Their reaction with CO<sub>2</sub> is slower but they require less energy for the desorption of CO<sub>2</sub>

and their CO<sub>2</sub> capture capacity is higher.<sup>[2,3]</sup> Numerous experimental studies the use of tertiary amines and their reactive properties for CO<sub>2</sub> capture exist.<sup>[4–13]</sup> Chowdhury et al.<sup>[4]</sup> screened 25 tertiary amine-based CO<sub>2</sub> absorbents for absorption/desorption rates, vapor-liquid equilibrium, and heat of reaction, revealing a correlation between the molecular structure of the amines and their efficiency for CO<sub>2</sub> absorption. Puxty et al.<sup>[6]</sup> screened 76 amines including tertiary ones and suggested that a stable intramolecular hydrogen bond that prevents carbamate formation, leads to improved CO<sub>2</sub> absorption. Xiao et al.<sup>[7]</sup> also investigated various commercial tertiary amines for CO<sub>2</sub> capture. They reported a correlation between the tertiary amine structures and the CO<sub>2</sub> absorption rate, CO<sub>2</sub> loading, pK<sub>a</sub>, and the heat of CO<sub>2</sub> absorption. They also suggested that the reactivity of tertiary amines for CO<sub>2</sub> capture is directly related to the properties of the side chains connected to the nitrogen atom. Bernhardsen et al.<sup>[9]</sup> reported that the number of hydroxyl groups in an amine molecule affects its reactivity for CO<sub>2</sub> capture; as the number of hydroxyl groups increases, the amine's basicity decreases, leading to reduced reactivity with CO<sub>2</sub>. This corresponds to the findings of Chowdhury et al. and Xiao et al. Linear chains in tertiary alkanolamines were also found to improve the CO<sub>2</sub> capture performance, as reported by Liu and coworkers.<sup>[11]</sup> Hartono et al.<sup>[13]</sup> evaluated 15 solvent candidates, including tertiary amines. They reported that TM-1,4-DAB, 1–2(HE)PRLD, and DEAE-EO are promising tertiary amine solvents with capacities comparable to MEA. From these studies one can conclude that tertiary amines are promising candidates for use as absorbents, and that their molecular structures are crucial for effective CO<sub>2</sub> capture.

Computational approaches have been used for examining and characterizing the reaction pathways of CO<sub>2</sub> capture in

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amines previously.<sup>[14–19]</sup> These tools can provide electronic properties and geometries of all species across the entire potential energy surface, which cannot be delivered solely by experiments. For example, Rozanska et al.<sup>[16]</sup> used quantum chemical calculations to develop a model for predicting the absorption rate of CO<sub>2</sub> in tertiary amines. Narimani et al.<sup>[17]</sup> used density functional theory to study the effect of substitution on tertiary amines for CO<sub>2</sub> capture. They reported that the performance of the tertiary amines studied by them for CO<sub>2</sub> absorption is comparable to that of methyldiethanolamine (MDEA), a conventional tertiary amine absorbent. A novel computational approach was employed by Orlov et al.<sup>[20]</sup> for identifying effective CO<sub>2</sub> capture solvents. They used machine learning to screen over 9000 solvents for CO<sub>2</sub> capture, evaluating factors like thermodynamic stability, viscosity, and mass transfer kinetics. However, the reaction mechanism and the characteristics of the potential energy surface in tertiary amine solutions have been rarely discussed in molecular detail. Especially the reaction pathway related to CO<sub>2</sub> capture has not yet been reported.

In this work, quantum chemical calculations based on density functional theory (DFT) have been employed to gain deeper insights into CO<sub>2</sub> capture by tertiary amines. We study the reaction mechanism and compare different amines. Then, we identify several DFT-based descriptors such as relative energies, geometries and electronic properties of the amines that influence the CO<sub>2</sub> loading. Lastly, we use the Sure-Independence Screening and Sparsifying Operator (SISSO) method<sup>[21]</sup> to provide a comprehensive mathematical expression for predicting the CO<sub>2</sub> loading.

## Results and Discussion

### CO<sub>2</sub> Capture Mechanism

The tertiary amines employed in this study were selected based on the experimental findings of M. Xiao et al.<sup>[7]</sup> These 10 molecules are depicted in Figure 1.

We investigated the base-catalyzed hydration mechanism that forms bicarbonate. This mechanism is widely accepted and has been extensively studied for CO<sub>2</sub> capture by tertiary amines.<sup>[14–18]</sup> The relative energies reported and discussed in this section refer to the typical temperature of 303.15 K. Results for the other temperatures are provided in the Tables S1–S3. First, we look at the enthalpy change ( $\Delta H$ ) associated with bicarbonate (HCO<sub>3</sub><sup>−</sup>) formation on DMEA via the reaction H<sub>2</sub>O +

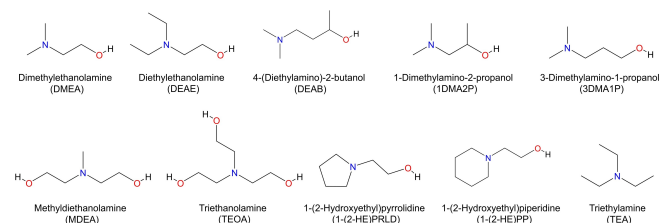


Figure 1. Tertiary amines employed in this study.

CO<sub>2</sub> → HCO<sub>3</sub><sup>−</sup>. This serves to validate the M06-2X functional used in our calculations, because in this relatively small system we can compare with results from the DLPNO-CCSD(T) method as well as with experimental values. DMEA is chosen for this purpose due to its acknowledged efficiency in CO<sub>2</sub> capture.<sup>[2]</sup> The  $\Delta H$  value from M06-2X is −12.9 kcal/mol, which is very close to the DLPNO-CCSD(T) value of −13.4 kcal/mol and the experimental range of −11.6 to −14.5 kcal/mol.<sup>[7,10]</sup>

The optimized structures of the reaction mechanism of the studied tertiary amines for CO<sub>2</sub> capture are shown in Figure S1 of the Supporting Information, with DMEA displayed as an example in Figure 2. The relative enthalpies for all amines are summarized in Figure 3. The reaction starts with the co-absorption complex between CO<sub>2</sub> and H<sub>2</sub>O molecules over the amines (Abs). In this complex, H<sub>2</sub>O interacts with the amine by forming a hydrogen bond between one of its hydrogen atoms (H) and the nitrogen atom (N) of the amine, while the CO<sub>2</sub> molecule is located in close proximity to the absorbed H<sub>2</sub>O. The structures of co-absorbed CO<sub>2</sub>, including both the C=O bond length and the O–C–O bond angle, differ slightly from those of the isolated molecule, due to the weak interaction with the absorbed H<sub>2</sub>O. As shown in Figure 3a, the co-absorption energies are in range of −2.9 to −7.9 kcal/mol. For the DMEA case it is −7.0 kcal/mol, which is slightly higher (more negative) than that of monoethanolamine (MEA) reported in previous theoretical studies (−5.7 kcal/mol).<sup>[14]</sup> Subsequently, this adduct

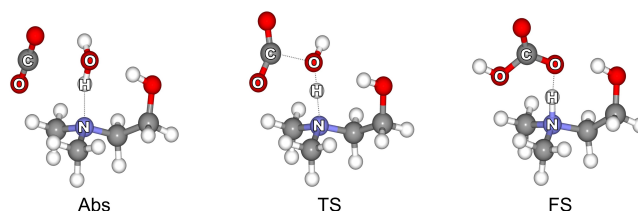


Figure 2. Reaction mechanism for CO<sub>2</sub> capture by DMEA via the base-catalyzed hydration mechanism.

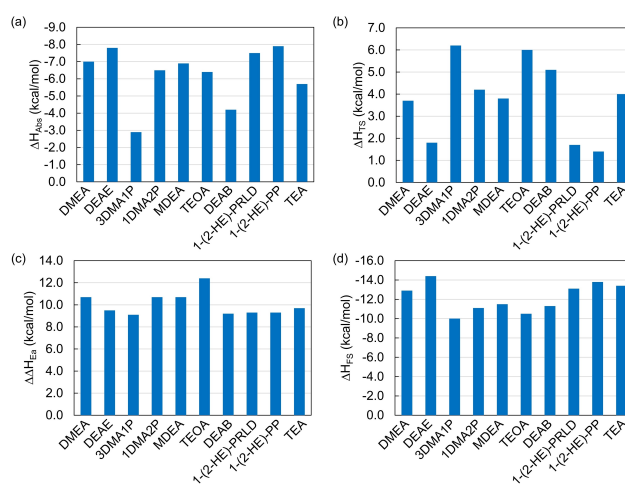


Figure 3. Relative enthalpies of (a) co-absorption ( $\Delta H_{Abs}$ ), (b) transition state energies ( $\Delta H_{TS}$ ), (c) activation barriers ( $\Delta\Delta H_{Ea}$ ) and (d) 3-C-hexonium intermediate energies ( $E_{FS}$ ) of CO<sub>2</sub> capture by amines via the base-catalyzed hydration mechanism.

is converted to bicarbonate via a transition state (TS), during which the proton (H) of H<sub>2</sub>O shifts to the nitrogen atom (N) of the amine. The oxygen (O) of H<sub>2</sub>O acts as a nucleophile and attacks the carbon atom (C) of CO<sub>2</sub> in a concerted manner (see Figure 2). This is facilitated by its geometry and electronic structure, notably the accessibility of the p\* anti-bonding orbital perpendicular to the molecular plane.<sup>[22]</sup> In the TS, the O–H bond of H<sub>2</sub>O is lengthened, while the N...H and C...O distances are shortened. The CO<sub>2</sub> molecule is also bent since hybridization of C changes from linear sp to trigonal planar sp<sup>2</sup>. The transition states possess one imaginary frequency, with normal modes connecting it to the adjacent minima listed in Table S4. The transition state energies of the ten tertiary amines range from 1.4 (1-(2-HE)-PP) to 6.2 kcal/mol (3DMA1P) and are in the order of 1-(2-HE)-PP < 1-(2-HE)-PRLD ≤ DEAE < DMEA ≤ MDEA ≤ TEA ≤ 1DMA2P < DEAB < TEOA ≤ 3DMA1P.

The activation barriers (E<sub>a</sub>), defined as the energy differences between the co-absorption state (Abs) and the transition state (TS), range from 9.1–12.4 kcal/mol. The activation barrier for DEAE is 9.5 kcal/mol, which closely matches a literature value of 9.7 kcal/mol obtained on the B3LYP/6-31G(d) level of theory.<sup>[19]</sup> For MDEA, the computed activation energy is 10.7 kcal/mol, consistent with experimental data (10.1 kcal/mol).<sup>[23]</sup> This value also aligns well with previous theoretical studies, which reported an activation energy of 9.8 kcal/mol using the B3LYP/6-311++G(d,p) level of theory.<sup>[17]</sup> Additionally, 1DMA2P yields an activation barrier close to the previous calculation using B3LYP with the def2-TZVPP basis set,<sup>[19]</sup> with values of 10.7 kcal/mol compared to 9.8 kcal/mol, respectively.

After the TS, the zwitterionic complex in which bicarbonate is bound to the protonated amine through hydrogen bonds, is formed (see FS in Figure 2). The CO<sub>2</sub> molecule contained in this intermediate exhibits a trigonal planar (sp<sup>2</sup>) hybridization. Similar structures have been found in carbamates in aqueous amines, bicarbonate in seawater, and in amine-functionalized solid sorbents that also capture CO<sub>2</sub> as carbamates.<sup>[22]</sup> The relative energies range from −10.1 (3DMA1P) to −14.4 kcal/mol (DEAE). The trend of this relative energies is found to be the same as that of the TS energies (see Figure 3d). The trend in ΔH of HCO<sub>3</sub><sup>−</sup> formation for molecules with the linear −CH<sub>2</sub>CH<sub>2</sub>−OH group (DEAE > DMEA > MDEA > TEOA) is in good agreement with experimental measurements.<sup>[10]</sup> Similarly, the trend observed for 1DMA2P and 3DMA1P is also consistent with experimental data.<sup>[10]</sup> For temperatures different to 303.15 K (298.15, 308.15, and 313.15 K) the relative orders stay the same in all states along the reaction coordinate (see Tables S1–S3).

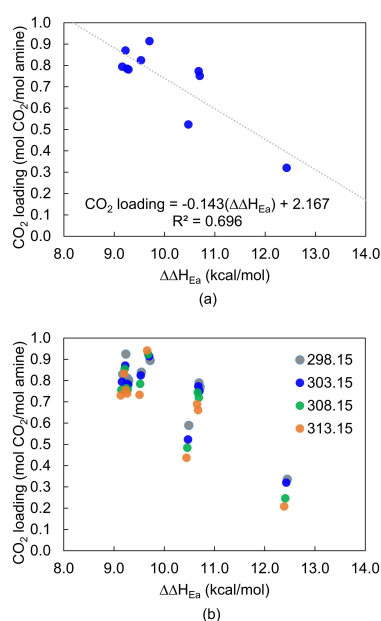
### Structure-Activity Relationships

To study the relationships between the molecular structures of amines and the activation energies for the conversion CO<sub>2</sub> to HCO<sub>3</sub><sup>−</sup>, we take into account the following features: the side chains (R) attached to the N atom, the chain length, the linear and branched configurations, the number of R-OH groups attached to the N atom, and the differences between cyclic and

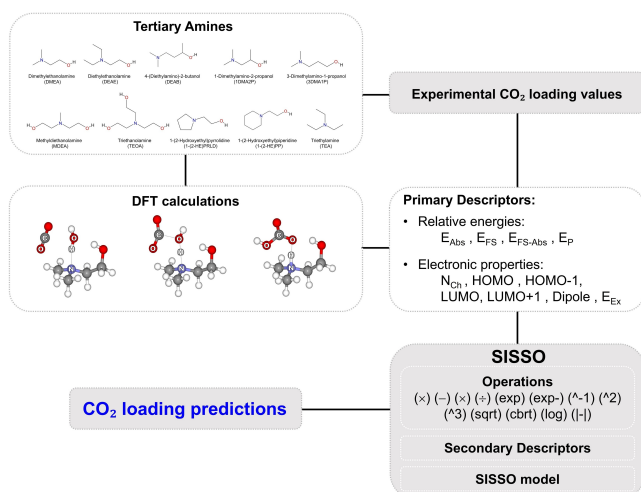
linear structures. We first compare DMEA where the side chains are methyl groups with DEAE where they are ethyl groups. As shown in Figure 3, the apparent (ΔH<sub>TS</sub>) and intrinsic activation energies (ΔΔH<sub>Ea</sub>) of DEAE are 1.8 and 9.5 kcal/mol, respectively, which are lower than in DMEA (3.7 and 10.7 kcal/mol). The trend in the barriers agrees well with the experimental CO<sub>2</sub> loading yield<sup>[7]</sup> and also with the rate constant (k<sub>0</sub>)<sup>[11]</sup> of the tertiary amine reported previously. The reason for the lower barrier in DEAE is likely the higher electron-donating capability of its ethyl groups compared to the methyl groups in DMEA. This results in a more negative charge on the N atom of DEAE (−0.627e) than that of DMEA (−0.612e). It makes sense that the charge on the amine nitrogen should be important in facilitating proton abstraction from the H<sub>2</sub>O molecule and the nucleophilic attack by OH<sup>−</sup> on CO<sub>2</sub>. When comparing DMEA with 3DMA1P, which contains even longer side chains, the apparent barrier of DMEA (3.7 kcal/mol) is lower than that of 3DMA1P (6.2 kcal/mol). However, the intrinsic barrier is 2 kcal/mol lower due to the less stable absorption complex (Abs) resulting from weaker hydrogen bonding. We also can compare linear and branched configurations of the R-OH groups (DMEA and 1DMA2P, respectively) and observe no difference in activation barriers. We can also look at the effect of the number of R-OH groups substituted on the N atom by comparing DMEA, which contains one hydroxyethyl group, with MDEA and TEOA, which contain two and three hydroxyethyl groups, respectively. The barriers for DMEA and MDEA do not change (see Figure 3). However, increased substitution with hydroxyethyl group results in higher barriers, both apparent and intrinsic, from 3.8 and 10.7 kcal/mol in the case of MDEA to 6.0 and 12.4 kcal/mol in TEOA. These barrier heights are again in line with CO<sub>2</sub> loading values reported in experiments.<sup>[7]</sup> The barrier differences might be due to increased steric hindrance, which increases with the number of hydroxyethyl groups, thereby reducing the reactivity of the N atoms. DMEA can also be compared with cyclic the structures 1-(2-HE)-PRLD and 1-(2-HE)-PP. The activation barrier for both 1-(2-HE)-PRLD and 1-(2-HE)-PP amines is 9.3 kcal/mol, which is lower than that of DMEA (10.7 kcal/mol), resulting in a greater activity of the cyclic structures. Altogether, it can be assumed that amines containing side chains with higher electron-donating group effects, such as the ethyl group in this study, along with a single R-OH group, and cyclic structures, exhibit good activity for CO<sub>2</sub> capture to HCO<sub>3</sub><sup>−</sup>. However, when comparing amines with and without an R-OH group, such as DEAE and TEA, the apparent barrier for DEAE is 3 kcal/mol lower than that for TEA (see Figure 3). The transition state (TS) in case of DEAE is stabilized by hydrogen bonds from the OH group whereas the one of TEA is not. Despite this, the intrinsic barriers remain the same since also the co-absorption state of TEA is also less stable than the one for DEAE. Thus, the presence of an R-OH group in the amine molecule stabilizes both the absorption and transition state complexes, resulting only in a shift of the potential energy pathway.

CO<sub>2</sub> Loading Prediction

The analysis of the energetic trends and the structure-activity relationships above suggests a correlation between the activation barriers and the CO<sub>2</sub> loading of amines. We look further into this relationship using the CO<sub>2</sub> loading values reported by M. Xiao et al.<sup>[7]</sup> In his work, a vapor-liquid equilibrium (VLE) apparatus was used to measure the equilibrium solubility of CO<sub>2</sub> in various tertiary amine-based absorbents. They introduced a constant volume of tertiary amines (20 mL of a 2 M solution) into the reactor. A mixture of CO<sub>2</sub> and N<sub>2</sub> was used as the inlet gas at atmospheric pressure, with the CO<sub>2</sub> partial pressure maintained at 15 kPa. The experiments were conducted at temperatures of 293, 298, 303, and 313 K. As shown in Figure 4a, we observe a modestly linear correlation between



**Figure 4.** Correlation of  $\Delta\Delta H_{Ea}$  and CO<sub>2</sub> loading for the temperature of 303.15 K (a) and all studied temperatures (b).



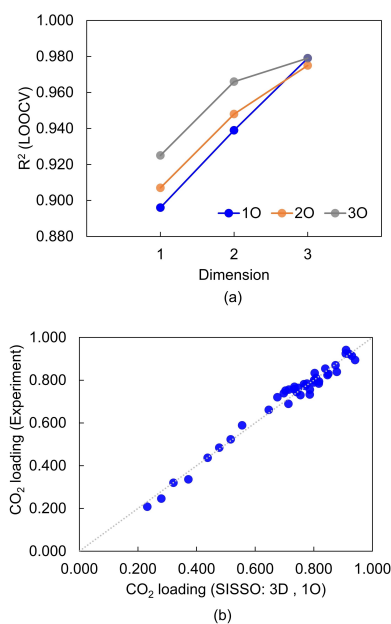
**Figure 5.** Workflow for CO<sub>2</sub> loading of amines prediction in the present work.

Ea and the CO<sub>2</sub> loading with an R<sup>2</sup> coefficient of 0.696 and a RMSE of 0.097 for the temperature of 303.15 K. In Figure 4b the other temperatures (298.15, 303.15, 308.15, and 313.15 K) are also plotted. Ea is therefore at least one suitable descriptor for predicting the CO<sub>2</sub> loading.

Figure 4a shows that the predictive power of predicting CO<sub>2</sub> loading by using Ea alone is still modest. Moreover, the determination of Ea still required to locate the TS structures. This is not always easy and often time consuming for non-specialists. An advantage would be to solely use descriptors which depend on the reactants. For this, we selected 12 primary descriptors derived from DFT calculations, including relative energies, geometries, and electronic properties of the amines. These were then transformed by means of mathematical expressions, thus creating a large set of secondary descriptors. The SISSO method<sup>[21]</sup> is then used to screen these terms and use a small subset for fitting towards experimental loading values. The workflow is shown in Figure 5. As primary descriptors we selected temperature (T), the relative enthalpy in the co-absorption complex ( $H_{Abs}$ ), bicarbonate formation complex ( $H_{FS}$ ), the energy difference between FS and Abs ( $H_{FS-Abs}$ ), and protonation energy of amines ( $E_P$ ), the parameters and properties defining the amines, such as dipole moment (D), charge of N atom ( $N_{Ch}$ ), highest occupied molecular orbital (HOMO;  $E_H$ ), lowest unoccupied molecular orbital (LUMO;  $E_L$ ), next-to-highest occupied molecular orbital (HOMO-1;  $E_{H-1}$ ), second lowest unoccupied molecular orbital (LUMO+1;  $E_{L+1}$ ) and first electronic excitation energy ( $E_{Ex}$ ). The SISSO method can evaluate a very large number of terms, here those generated by combining the primary descriptors revealed above in arbitrary mathematical expressions. The dimension (D), defined as the number of non-zero coefficients, and the number of allowed mathematical operations (O) are two parameters used to control model complexity and generate secondary descriptors. In this work, we consider complexities ranging from 1D,10 to 3D,30, resulting in a total of 9 model levels.

The comprehensive mathematical equations derived from all model complexities are given in Table S5. The determination coefficient ( $R^2$ ) and the RMSE of the trained models are shown in Figure S2. We notice that the protonation energy appears in all predictor equations (see Table S5), highlighting the significance of this descriptor. Obviously, the prediction accuracy of the trained model increases with complexity. The robustness of these predictions must be validated to avoid overfitting in highly complex models. We used the leave-one-group-out cross validation (LOOCV) procedure.<sup>[24]</sup> This procedure has been demonstrated to be suitable for small datasets, as present in this study.<sup>[25]</sup> We perform the analysis using 40 datasets obtained from all 10 amines across four temperatures. A model is trained using data from 39 sets and subsequently tested using the remaining 40<sup>th</sup> dataset. The overall cross validation performance is evaluated by  $R^2$  (see Figure 6a) and the RMSE (see Figure S3) between the predicted values and the actual values. The high  $R^2$  (0.896–0.979) and low RMSE (0.026–0.059) indicate that all SISSO models for predicting CO<sub>2</sub> loading are reasonably robust against overfitting. The SISSO model with 3D,10 model provides the  $R^2$  of 0.979 and the RMSE for the test





**Figure 6.**  $R^2$  of trained models after leave-one-group-out cross validation for different model complexities (a) and fit quality of according to Equation (1) where the complexity 3D and 1D had been selected (b).

set of 0.027 mol  $\text{CO}_2$ /mol amine which are comparable to the 3D,3D model ( $R^2$  of 0.979 and  $\text{RMSE}$  = 0.026 mol  $\text{CO}_2$ /mol amine) as shown in Figure 6a. Therefore, we suggest that the 3D, 1D model is the optimal model due to its lower complexity, making it more practical and less time-consuming for predictions. The SISSO with 3D and 1D complexity is given by the expressions shown in Equation (1).

$$\begin{aligned} \text{CO}_2 \text{ loading} = & -0.0017472(T \times D) \\ & + 0.0652473(N_{\text{Ch}} \times E_{\text{H-1}}) - 0.00029659 \\ & (E_{\text{H-1}} \times E_{\text{p}}) + 11.660927 \end{aligned} \quad (1)$$

The fitting result from Equation (1) is shown Figure 6b. Applying Equation (1) simplifies the number of DFT calculations needed by eliminating the calculations of absorption and intermediate complexes, as well as of TS structures. The predicted  $\text{CO}_2$  loadings specifically require only calculations on the isolated amine molecule. We note that various descriptors may be correlated with  $\text{CO}_2$  loading and could have predictive performance similar to the proposed descriptors. Moreover, expanding the database of amine structures could enhance predictive accuracy. Therefore, we plan to continue this work by utilizing larger databases, including both amine structures and their corresponding descriptors.

We further estimate the predictive ability of our SISSO descriptors in Equation (1) for  $\text{CO}_2$  loading on 1-diethylamino-2-propanol (DEA2P), which is not present in the dataset, as an external test. The optimized structures for the reaction on DEA2P are shown in Figure S4. Compared to the experimental  $\text{CO}_2$  loading (0.867 mol  $\text{CO}_2$ /mol amine<sup>[10]</sup>), the absolute error of the SISSO prediction (0.847 mol  $\text{CO}_2$ /mol amine) is 0.020 mol

$\text{CO}_2$ /mol amine. This confirms that the SISSO descriptors are robust and transferrable.

## Conclusions

We used DFT calculations to study the reaction mechanism and structure-activity relationships of tertiary amines in  $\text{CO}_2$  capture. We find that the  $\Delta H$  of bicarbonate formation on DMEA from DFT calculations is in good agreement with both DLPNO-CCSD(T) calculations and experimental data. The base-catalyzed hydration reaction mechanism for  $\text{CO}_2$  capture is investigated. It proceeds in a single step involving proton transfer and the formation of a carbon-oxygen bond to form bicarbonate. Tertiary amines containing ethyl group substitutions, a single hydroxyl group, and cyclic structures are found to exhibit promising activity for  $\text{CO}_2$  capture based on the reaction barrier analysis. We also observe a reasonable correlation between  $E_{\text{a}}$  and  $\text{CO}_2$  loading. Finally, we propose a workflow to derive a mathematical expression for predicting the experimental  $\text{CO}_2$  loading using only properties of the amines. This workflow uses the statistical learning technique SISSO to find the best-performing combination of DFT-based descriptors. We suggest a model which includes temperature, dipole moment, charge of N atom, HOMO-1, and protonation energy descriptors. This model of 3D and 1D complexity is optimal, based on the analyzed  $R^2$  and  $\text{RMSE}$  results after testing for robustness via cross-validation.

## Computational Methods

All calculations were performed using DFT with the M06-2X density functional<sup>[26–29]</sup> as implemented in Gaussian 16 Revision B.01.<sup>[30]</sup> This method takes into account medium-range dispersion interactions in its parameterization and provides high-quality results as has been shown in a large number of studies on the absorption and mechanism of several organic reactions.<sup>[31–35]</sup> The def2-SVP basis set was used to perform the geometric optimization. The energetics were further improved by performing single point calculations on the M06-2X optimized geometries using the def2-QZVP basis set. For a prototypical reaction the energetics were checked with CCSD(T) calculations in the complete basis set (CBS) extrapolation scheme as implemented in the ORCA program<sup>[36]</sup> using the DLPNO scheme. The polarizable continuum model (PCM)<sup>[37]</sup> was employed to model the solvation by water. Partial charges and orbital energies were analyzed with the natural bond orbital (NBO) method.<sup>[38]</sup> To identify the nature of all stationary points, vibrational frequency calculations were performed on the optimized geometries. Thermal corrections were also obtained from the frequency calculations. The relative energies in this work are reported as enthalpies ( $\Delta H$ ) at four temperatures (298.15, 303.15, 308.15, and 313.15 K) with a pressure of 1 atm. Thermal corrections were also obtained from the frequency calculations. The enthalpy ( $H$ ) is calculated according to Equation (2).

$$H = E_{\text{el}} + E_{\text{tot}} + k_{\text{B}}T \quad (2)$$

where  $E_{\text{el}}$  is the total electronic energy,  $E_{\text{tot}}$  is the internal thermal energy which contains the electronic, translational, rotational, and vibrational contributions,  $k_{\text{B}}$  is Boltzmann's constant, and  $T$  is

the temperature. The relative enthalpy change in the reaction is defined as follows:

$$\Delta H = H_{\text{complex}} - H_{\text{amine}} - H_{\text{CO}_2} - H_{\text{H}_2\text{O}} \quad (3)$$

where  $H_{\text{complex}}$  represents the enthalpy of the absorption, transition and intermediate states in the reaction coordinate,  $H_{\text{amine}}$  is the enthalpy of the amine molecule,  $H_{\text{CO}_2}$  is the enthalpy of the  $\text{CO}_2$  molecule and  $H_{\text{H}_2\text{O}}$  denotes the enthalpy of the  $\text{H}_2\text{O}$  molecule. Cartesian coordinates of all optimized structures are provided in the Supporting Information.

## Supporting Information Summary

Relative energies for  $\text{CO}_2$  capture by amines via the base-catalyzed hydration mechanism calculated at 298.15, 308.15 and 313.15 K (Tables S1–S3), optimized structured for the reaction of  $\text{CO}_2$  capture by the studied amines (Figure S1), imaginary frequency for the transition state structures (Table S4), SISSO model for  $\text{CO}_2$  loading prediction with various model complexities (Table S5),  $R^2$  (a) and the RMSE (b) of the training models (Figure S2), RMSE of the leave-one-group-out cross validation (Figure S3) optimized structured for the reaction of  $\text{CO}_2$  capture by DEA2P (Figure S4) and cartesian coordinates of all optimized structures.

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## Conflict of Interests

The authors declare no conflict of interest.

## Data Availability Statement

The data that support the findings of this study are available in the supplementary material of this article.

**Keywords:**  $\text{CO}_2$  capture · Amine · DFT · Machine learning · SISSO

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